Which of the following statements is correct?

A. A reference is useful when we intend to change the values of actual arguments through the function call.
B. A reference is useful when we want to save memory by preventing the creation of large structure variables that are being passed to the function.
C. Only A is correct.
D. Neither A nor B is correct.

Answer: A

Which of the following statements are incorrect related to malloc() function?

A. malloc() calls constructors automatically.
B. malloc() returns a void* pointer.
C. malloc() allocates raw memory.
D. malloc() initializes allocated memory to zero. Garbage

Answer: A or A & D

What will be the output of the following C++ code?

```
#include<iostream>
#include<stdlib.h>
using namespace std;

class Test
{
    int numI;
public:
    Test()
    {
        cout << "Constructor called";
    }
};

int main()
{
    Test *t = (Test *) malloc(sizeof(Test));
    return 0;
}
```

Class Test // empty class
{
};
1 byte

4 byte
4 byte raw

A. Constructor called
B. Compilation Error
C. Runtime Error
D. No Output

Answer: D

A destructor takes _____ arguments.

A. One
B. Two
C. Any number of arguments
D. Zero

Answer: D

What is the difference between delete and delete[] in C++?

A. delete is used for a single object, while delete[] is used for an array of objects.
B. delete[] calls the destructor for each element of the array.
C. delete calls the destructor only once.
D. All of the above.

Answer:

Which operator is used to allocate memory dynamically in C++?

A. malloc
B. alloc
C. new
D. create

Answer:

Which operator is used to deallocate memory allocated using new?

A. free()
B. delete
C. release()
D. destroy()

Answer:

What is the main advantage of passing arguments by reference?

A. Increases memory usage
B. Prevents modification of original data
C. Avoids copying and improves performance
D. Makes variables global

Answer:

Which of the following is true about a reference variable?

A. It can be NULL.
B. It must be initialized when declared.
C. It can be reseated to another variable later.
D. It occupies no memory.

Answer:

Which memory area stores dynamically allocated objects?

A. Stack
B. Code Segment
C. Heap
D. Data Segment

Answer: C

Wrapping of private data in classes in object-oriented programming languages is termed as

A. Inheritance
B. Encapsulation
C. Polymorphism
D. Abstraction

Answer

Function Overloading comes under?

A. Runtime Polymorphism
B. Compile-time Polymorphism
C. Inheritance
D. Encapsulation

Answer: B

Minor pillars of object-oriented programming language?

I. Polymorphism
II. Concurrency
III. Persistence
IV. Hierarchy

A. I and II only
B. II, III and IV only
C. I, II, III and IV
D. III and IV only

Answer:

_____ always describes the outer behavior of an object.

A. Encapsulation inner behavior
B. Inheritance
C. Abstraction
D. Constructor

Answer: C

Which feature can be implemented using encapsulation?

A. Data Hiding
B. Function Overloading
C. Inheritance
D. Dynamic Binding

Answer:

Which feature can be implemented using encapsulation?

A. Data Hiding
B. Function Overloading
C. Inheritance
D. Dynamic Binding

Answer:

Which of the following is known as Data Hiding?

A. Declaring data members as public
B. Declaring data members as private and accessing them through member functions
C. Using global variables
D. Using friend functions

Answer:

Which OOP concept allows one class to acquire properties of another class?

A. Encapsulation
B. Abstraction
C. Inheritance
D. Polymorphism

Answer:

Which OOP concept allows the same function name to perform different tasks?

A. Encapsulation
B. Polymorphism
C. Abstraction
D. Aggregation

Answer:

Which access specifier makes class members accessible only within the class?

A. public
B. protected
C. private
D. friend

Answer:

Which of the following best defines Abstraction?

A. Hiding implementation details and showing only essential features
B. Acquiring properties from another class
C. Binding data and methods together
D. Using multiple constructors

Answer:

What is the effect of declaring a data member as const in
A. It can be modified after initialization
B. It cannot be modified after initialization
C. It is shared between objects of a class
D. It is mutable
Answer: B

What is the effect of declaring a member function as const ?
A. It can modify the state of an object
B. It cannot modify the state of an object
C. It is shared between objects of a class
D. It is static
Answer:

What is the purpose of the mutable keyword in C++?

A. To make data members constant Const
B. To make data members static static
C. To allow modification of non constant data members in const member function
D. To hide data members from outside access → privale

What is the benefit of using a modular approach in programming?
A. It makes the code more complex
B. It makes the code more difficult to maintain
C. It allows for code reuse and easier maintenance
D. It reduces the performance of the program

Can a constructor function be constant?
A. Yes, a constructor can be declared as const.
B. It depends on the programming language being used.
C. No, a constructor cannot be declared as const because it needs to initialize the object's members.
D. Constructors can be const, but only for static objects.
C

How are the constants declared?
A. const keyword
B. #define preprocessor
C. both const keyword and #define preprocessor
D. $define
Answer:

function defn → encapsu
function call ⟺ abstraction